

Week 3

- **Problem 1**

We say that two integers have *different parity* if one of them is odd and the other one is even. Prove that two integers have different parity if and only if their sum is odd.

- **Proof 1.1**

($\Rightarrow$) Suppose two integers have different parity. Without loss of generality, let one of them be $2k_1$ for some $k_1 \in \mathbb{Z}$ and the other one be $2k_2 + 1$ for some $k_2 \in \mathbb{Z}$. Then their sum

$$2k_1 + (2k_2 + 1) = 2(k_1 + k_2) + 1$$

is odd because $k_1 + k_2 \in \mathbb{Z}$.

($\Leftarrow$) Let the two integers be a and b , and suppose that their sum is odd, that is, $a + b = 2s + 1$ for some $s \in \mathbb{Z}$. By definition, the integer a must be either even or odd. We consider these two cases:

1. **Case 1:** Suppose a is even, so $a = 2k_1$ for some $k_1 \in \mathbb{Z}$.

Then we can express b as:

$$b = (a + b) - a = (2s + 1) - 2k_1 = 2(s - k_1) + 1.$$

Since $s - k_1 \in \mathbb{Z}$, b is an odd integer. Thus, a is even and b is odd, meaning they have different parity.

2. **Case 2:** Suppose a is odd, so $a = 2k_1 + 1$ for some $k_1 \in \mathbb{Z}$.

Then we can express b as:

$$b = (a + b) - a = (2s + 1) - (2k_1 + 1) = 2(s - k_1).$$

Since $s - k_1 \in \mathbb{Z}$, b is an even integer. Thus, a is odd and b is even, meaning they have different parity.

Therefore, in either case, the two integers must have different parity.

- **Problem 2**

Suppose that n lines are drawn in the plane so that

1. no two lines are parallel;
2. no three lines pass through the same point.

Prove by induction that the lines divide the plane into

$$1 + \frac{n(n+1)}{2}$$

regions.

- **Proof 2.1**

By induction.

1. Base step

If $n = 1$ the result is true since the number of regions is $1 + \frac{1 \cdot (1+1)}{2} = 2$.

2. Inductive step

Suppose that the result is true for n , that is, any n lines satisfying the conditions divide the plane into

$$1 + \frac{n(n+1)}{2}$$

regions.

Now consider the case with $n + 1$ lines. When we add the $(n + 1)$ -th line to the existing n lines:

- Since the new line is not parallel to any of the existing n lines, it must intersect all n lines at n distinct points.
- These n intersection points divide the new $(n + 1)$ -th line into exactly $n + 1$ line segments including rays at the ends.
- Each of these $n + 1$ segments cuts through an existing region and splits it into two, then creating exactly $n + 1$ new regions.

Hence, the total number of regions for $n + 1$ lines is

$$\begin{aligned} \left(1 + \frac{n(n+1)}{2}\right) + (n+1) &= 1 + \frac{n(n+1) + 2(n+1)}{2} \\ &= 1 + \frac{(n+1)(n+2)}{2} \\ &= 1 + \frac{(n+1)((n+1)+1)}{2}. \end{aligned}$$

Then the result is true for $n + 1$ and therefore the lines divide the plane into $1 + \frac{n(n+1)}{2}$ regions for every $n \in \mathbb{N}$.

• **Problem 3**

Prove by induction that for every $n \in \mathbb{N}_0$,

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4).$$

◦ **Proof 3.1**

By induction.

1. Base step

If $n = 0$ then the result is true because $5^{0+1} + 3 \cdot 9^0 - 4 = 4$ and $4 \mid 4$.

2. Inductive step

Suppose that the result is true for n , that is,

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4).$$

By definition, $\exists k \in \mathbb{Z}, 5^{n+1} + 3 \cdot 9^n - 4 = 4k \implies 5^{n+1} + 3 \cdot 9^n = 4(k+1)$. Then

$$\begin{aligned} 5^{(n+1)+1} + 3 \cdot 9^{n+1} - 4 &= 5 \cdot 5^{n+1} + 9 \cdot 3 \cdot 9^n - 4 \\ &= 5 \cdot (5^{n+1} + 3 \cdot 9^n) + 4 \cdot 3 \cdot 9^n - 4 \\ &= 5 \cdot 4(k+1) + 4 \cdot 3 \cdot 9^n - 4 \\ &= 4 \cdot (5k + 4 + 3 \cdot 9^n). \end{aligned}$$

Since $5k + 4 + 3 \cdot 9^n \in \mathbb{Z}$, we have $4 \mid (5^{(n+1)+1} + 3 \cdot 9^{n+1} - 4)$. Then the result is true for $n + 1$ and therefore

$$4 \mid (5^{n+1} + 3 \cdot 9^n - 4)$$

is true for every $n \in \mathbb{N}_0$.